

BONGIMUSA SANDILE KHOZA

Software Developer ■ AI & Machine Learning Engineer ■ Data Analyst ■ Cybersecurity & IT Risk ■ QA & Testing

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Citizenship: South African | Driver's Licence: Code 10 (C1) | Availability: Immediately Available | Willing to Relocate: Yes

PROFESSIONAL SUMMARY

Results-driven BSc Computer Science and Mathematics graduate (NQF Level 7, North-West University, conferred May 2026) with a broad, practical technology foundation spanning software development, artificial intelligence, machine learning, data analytics, cybersecurity, IT risk and compliance, QA and testing, and network infrastructure. Has independently designed, built, and shipped **two live AI-powered applications** — a facial recognition ride safety platform and an LLM-powered email automation tool — demonstrating the ability to take ideas from concept to production. Combines strong quantitative thinking, attention to detail, clear communication, and genuine passion for technology that solves real problems. South African citizen, immediately available, willing to relocate.

TECHNICAL SKILLS

Languages & Development: Python, JavaScript, Java, C++, HTML5, CSS3, SQL, Node.js, Streamlit — OOP, Data Structures & Algorithms, REST APIs, SDLC, Agile/Scrum

AI, ML & Computer Vision: Scikit-learn, DeepFace, KMeans Clustering, Neural Networks, Deep Learning, NLP, Computer Vision, Face Recognition, TextBlob

LLMs & Generative AI: LLaMA 3.3 70B (Groq API), Gemini 2.5 Flash (Gemini API), Claude AI, Prompt Engineering, Azure AI Engineer, Azure Cognitive Services

Data Analytics & BI: Pandas, NumPy, Matplotlib, Plotly, Power BI, MS Excel (VLOOKUP, Pivot Tables, Macros, Dynamic Dashboards), SQL, Statistical Analysis, Financial Modelling

QA, Testing & IT Risk: Functional & System Testing, Test Plan & Test Case Design, Defect Tracking, Root Cause Analysis, IT Risk Assessment, Control Gap Identification, Audit Documentation, BABOK Awareness

Cybersecurity & Compliance: Cybersecurity Fundamentals, POPIA, RBAC, Network Security, DHCP Snooping, Security Testing, Incident Management, ISO 9001 / COBIT / NIST Awareness

Networking & Infrastructure: Cisco Packet Tracer, VLANs, DHCP (IPv4/IPv6), Static NAT, TCP/IP, Protocol Analysis, Inter-VLAN Routing, Linux (Basic), System Performance Monitoring

DevOps & Tools: Git/GitHub, VS Code, Jupyter Notebook, Google Colab, Streamlit Cloud, Render, GitHub Pages, MS Office 365

LIVE DEPLOYED PROJECTS

SafeRide — AI Face Recognition Ride Safety Platform

Python • Streamlit • DeepFace • Gemini 2.5 Flash • PostgreSQL • Streamlit Cloud | May 2026

<https://saferide-rwv43c8efmxhtbmu2xgjr.streamlit.app> | github.com/SANDILE19991111/saferide

- Engineered a production-grade AI safety platform using DeepFace computer vision for live biometric passenger identity verification — directly applicable to cybersecurity, IAM, and fintech security domains
- Built end-to-end features: face recognition sign-up/sign-in with confidence scoring, dynamic ride pricing (R12–R25/km with peak surcharges), SOS emergency location logging, AI safety briefings via Gemini 2.5 Flash, and a full ride history dashboard
- Resolved complex real-world deployment challenges: DeepFace installation on cloud, secure API key management via Streamlit Secrets, camera support configuration, and button state management
- Demonstrates full-stack AI product development — from model integration to cloud deployment — with measurable real-world social impact

Email Drafter — LLM-Powered AI Automation Tool

Node.js • Express.js • Groq API • LLaMA 3.3 70B • JavaScript • Render | May 2026

<https://email-drafter-ciay.onrender.com> | github.com/SANDILE19991111/email-drafter

- Built and shipped a full-stack AI email generation tool integrating LLaMA 3.3 70B via Groq API — supporting multiple tones (Professional, Friendly, Casual, Persuasive) with AI-generated subject lines
- Implemented production-grade system prompt engineering to govern AI behaviour, secure API key handling, copy-to-clipboard UX, and automated CI/CD deployment pipeline on Render
- Demonstrates backend API integration, prompt engineering discipline, and production deployment skills

ACADEMIC & BOOTCAMP PROJECTS

Clinic Appointment System — Full-Stack Agile Delivery & QA

JavaScript • SQL • PostgreSQL • Agile | Apr – May 2025

- Designed a 4-entity normalised relational database (ERDs, SQL DDL/DML) and built a JavaScript frontend with RBAC and POPIA-compliant security controls
- Executed 15+ functional test cases, identified 5+ control gaps, and produced audit-style working papers — demonstrating QA and IT risk competency
- Delivered on time within an Agile team of 4; facilitated an exit-style stakeholder review presenting findings, risks, and corrective recommendations

Enterprise Network Infrastructure Design

Cisco Packet Tracer • VLANs • DHCP • NAT • TCP/IP | Aug – Oct 2025

- Designed 5 network topologies for a simulated enterprise; diagnosed 3 critical security failures through root cause analysis — restored 100% connectivity
- Produced full audit and configuration documentation covering VLAN segmentation, inter-VLAN routing, DHCP (IPv4/IPv6) with snooping, and Static NAT

KMeans Clustering — Machine Learning Data Pipeline

Python • Scikit-learn • NumPy • Matplotlib | Mar – May 2025

- Analysed 300+ user dataset; achieved 98% clustering accuracy (ARI: 0.3440) — produced 4 visualisations and a structured analytical report with actionable insights

Image Classification — Neural Network Model

Neural Networks • Google Teachable Machine | Feb 2025

- Trained a neural network on 100+ images across 2 classes; evaluated generalisation on unseen data; proposed retail automation application for the model

Database, Budgeting & Business Planning

MS Excel • SQL • ERD | Apr – Jun 2022

- Automated a payroll database for 15+ staff with zero discrepancies; built a 12-month dynamic financial budget dashboard and a procurement proposal with cost analysis, asset listing, and Gantt scheduling

EDUCATION

Bachelor of Science: Computer Science and Mathematics — NQF Level 7

North-West University, Mafikeng | Jan 2022 – Jan 2025 | Conferred: 5 May 2026 | SAQA ID: 73059

- Core modules: Artificial Intelligence, Machine Learning, Databases, Software Engineering, Data Structures & Algorithms, OOP, Computer Networks, Operating Systems, Cybersecurity Fundamentals, Statistics, Linear Algebra, Calculus
- Languages applied: Python, Java, C++, JavaScript, SQL — across AI, systems, data, and network projects
- Completed full SDLC project lifecycles and Agile team delivery with documentation, sprint planning, and stakeholder reporting

National Certificate: N3 — Engineering Studies | Ekurhuleni West College | 2019

National Senior Certificate — Bachelor's Pass | Mchaka High School | Dec 2018

CERTIFICATIONS & PROFESSIONAL TRAINING

Full Stack Development Certificate	FNB App Academy	Jul 2025
Generative AI: Prompt Engineering Basics	IBM / Coursera	May 2026
Introduction to Artificial Intelligence	IBM / Coursera	May 2026
AI For Everyone	DeepLearning.AI / Coursera	Apr 2026
Introduction to Generative AI	Google Cloud / Coursera	May 2026
Python for Data Science, AI & Dev	IBM / Coursera	2026
AI Essentials	Coursera	Jun 2026

ADDITIONAL INFORMATION

Citizenship: South African (African)	Date of Birth: 11 November 1999 (Age 26)
Driver's Licence: Code 10 (C1) — valid	Languages: English (Fluent), Xitsonga (Home Language), IsiZulu (Conversational)
Employment Status: Currently unemployed — immediately available	Willing to Relocate: Yes
Criminal Record: None	

REFERENCES

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